

DevTrack

A Mission-Driven Transparency Platform

Complete Feature Documentation

Platform: Kathmandu Metropolitan City (KMC) — Beta v1.0

Audience: Citizens, Officials, Experts, Administrators

Status: Public Beta — Live Tonight

Built by: Team DevTrack — Incubate Nepal 2024

Aligned with: Open Government Partnership (OGP) Framework

Table of Contents

1. Executive Summary.....	1
2. The Problem DevTrack Solves.....	2
2.1 Symptoms of the Transparency Gap.....	2
2.2 Why Existing Solutions Fall Short.....	3
3. The DevTrack Solution.....	3
3.1 The Four Pillars.....	3
Pillar 1 — Transparency.....	3
Pillar 2 — Accountability.....	4
Pillar 3 — Participation.....	4
Pillar 4 — Trust.....	4
4. Platform Architecture.....	4
4.1 Technology Stack.....	5
4.2 Data Model Overview.....	5
4.3 API Surface.....	5
5. User Roles and Permissions.....	5
5.1 Demo Accounts (Beta).....	6
6. Home Page — The Mission Front Door.....	6
6.1 Hero Section.....	6
6.2 Mission Pillars Section.....	7
6.3 Featured Projects.....	7
6.4 Live Feed — Latest Project Updates.....	7
6.5 Eight-Sector Grid.....	7
6.6 Beta Call-to-Action.....	8
7. Projects Explorer.....	8
7.1 Filtering and Search.....	8
7.2 Project Card.....	9
7.3 Status Categories.....	9
8. Project Detail Page.....	9

8.1 Project Header.....	9
8.2 Stats Row.....	10
8.3 About This Project.....	10
8.4 Budget Breakdown.....	10
8.5 Timeline and Milestones.....	10
8.6 Project Updates.....	11
8.7 Project Facts Sidebar.....	11
8.8 Location Preview.....	11
8.9 Public Discussion Link.....	11
9. Public Forum.....	12
9.1 Thread Structure.....	12
9.2 Comments and Replies.....	12
9.3 Voting.....	12
9.4 Creating a New Thread.....	13
9.5 Forum Examples in the Beta.....	13
10. Officials Directory — Direct Contact.....	13
10.1 Search and Filter.....	14
10.2 Official Card.....	14
10.3 Officials in the Beta.....	14
11. Citizen Dashboard.....	14
11.1 Profile Header.....	15
11.2 Tracked Projects.....	15
11.3 My Ward.....	15
11.4 Notifications.....	15
11.5 Quick Actions.....	16
12. Beta Feedback Wall.....	16
12.1 Submit Form.....	16
12.2 Feedback Categories.....	16
12.3 Feedback Wall.....	17
12.4 Average Rating.....	17
13. About / Vision Page.....	17

13.1 Mission Statement.....	17
13.2 Problem, Solution, Frameworks, Beyond Kathmandu.....	18
13.3 Six Goals.....	18
13.4 The Team.....	18
14. Design System.....	19
14.1 Color Palette.....	19
14.2 Typography.....	19
14.3 Visual Effects.....	19
14.4 Logo.....	20
15. Seeded Project Portfolio.....	21
15.1 Total Platform Metrics.....	21
16. Roadmap — What's Next for DevTrack.....	21
16.1 v1.1 — Immediate Priorities (Next 30 Days).....	21
16.2 v1.2 — Data and Verification (60–90 Days).....	22
16.3 v2.0 — Scale and Integration (6–12 Months).....	23
17. Technical Implementation Details.....	24
17.1 File Structure.....	24
17.2 Database Seeding.....	24
17.3 Authentication Flow.....	24
17.4 Performance.....	24
17.5 Responsive Design.....	25
18. Alignment with Global Frameworks.....	25
18.1 Open Government Partnership (OGP).....	25
18.2 Nepal e-Government Interoperability Framework (NeGIF).....	25
18.3 World Bank Digital Dividends (2016).....	26
19. Conclusion.....	26

Note: Right-click the table of contents above and choose "Update Field" to refresh page numbers.

1. Executive Summary

DevTrack is a mission-driven transparency and accountability platform built for Kathmandu Metropolitan City (KMC) and designed to scale across all 753 local governments of Nepal. The platform was conceived by six student researchers during the Incubate Nepal 2024 program, with the goal of ending the communication gap between government and citizens by making every development project — its budget, schedule, contractor, and accountable officials — visible, discussable, and accountable to the people it serves.

At its core, DevTrack is a web application that brings total transparency to government projects across Kathmandu Valley. Citizens can track projects in their ward, debate them on a public forum, vote on each other's comments, and reach elected officials directly through publicly listed contact information. Expert voices are prioritized through a verified badge system, ensuring that KMC receives high-quality feedback from domain specialists while still amplifying every citizen's voice.

This document is the complete reference for every feature in the DevTrack platform. It is intended for citizens learning how to use DevTrack, government officials considering adoption, journalists investigating project data, developers extending the platform, and stakeholders evaluating DevTrack for Open Government Partnership (OGP) alignment. Each section explains what a feature does, who it serves, how it works, and why it matters for the broader mission of good governance in Nepal.

The DevTrack Mission

DevTrack exists so that every Nepali — in Kathmandu today, in all of Nepal tomorrow — can see what their government is doing, question it, and shape it. We build it in the open, with citizens, for citizens. — Team DevTrack, Incubate Nepal 2024

2. The Problem DevTrack Solves

Kathmandu Valley is experiencing rapid urbanization, with hundreds of development projects — road construction, drainage, schools, hospitals, river cleanup, public toilets, smart parking, solar installations, disaster preparedness systems — running simultaneously across the metropolitan area. Yet citizens have no single place to see what is being built, who is building it, how much it costs, when it will finish, or who to hold accountable when things go wrong.

2.1 Symptoms of the Transparency Gap

- Budget overruns and delays that no one can predict or prevent because there is no public timeline to compare against.
- Contractors operating without public scrutiny, leading to inconsistent quality and missed deadlines.
- Officials who are nominally accountable but practically unreachable — no public phone, no public email, no surgery hours, no direct channel.
- Citizens who feel powerless because their concerns about noise, dust, flooding, or service quality have nowhere to go.
- A trust deficit between KMC and the people it serves, which makes every new project harder to deliver because citizens assume the worst.

2.2 Why Existing Solutions Fall Short

Nepal has made real progress on e-governance — the Public Procurement Monitoring Office (PPMO) has an online system, the Nepal e-Government Interoperability Framework (NeGIF) defines standards, and the Nagrik App provides citizen services. But none of these give citizens a single, plain-language view of every development project in their ward, with budgets, timelines, contractors, accountable officials, and a public discussion forum all in one place. DevTrack fills that gap.

3. The DevTrack Solution

DevTrack is built on four pillars: Transparency, Accountability, Participation, and Trust. Every feature in the platform maps to at least one of these pillars. The platform organizes every KMC development project into one of eight categories — Infrastructure, Education, Healthcare, Economy, Environment, Policy, Urban Development, and Citizen Benefits — making it easy for any citizen to find the projects that matter to them.

3.1 The Four Pillars

Pillar 1 — Transparency

Every KMC project is published in one place, in plain language, with full budget breakdowns by line item (materials, labor, equipment, overhead), milestone timelines with due dates and completion dates, named contractors and consultants, funding sources, and ward locations. Citizens do not need to file Right to Information requests — the information is already public, already searchable, and already categorized.

Pillar 2 — Accountability

Every project has a named owner — a ward chair, a department chief, or the Mayor's office. Every milestone has a due date, and when a milestone becomes overdue, it is flagged in red on the project page. Every delay is visible. Every budget overrun is visible. Citizens can hold specific people accountable, not just an abstract "the government."

Pillar 3 — Participation

A public forum is attached to every project, plus a city-wide forum for general discussion. Any citizen can post concerns, questions, or suggestions. Expert citizens (verified engineers, doctors, urban planners, environmental scientists) get a green Expert badge so their input is prioritized. Officials get a blue Official Response badge so citizens can see when their government has actually replied. Citizens can upvote and downvote comments to surface the most useful contributions.

Pillar 4 — Trust

When citizens can see what their government is doing, when they can ask questions and get real answers, when they can track projects over months and see them actually complete — trust grows. DevTrack is not a tool for shaming government; it is a tool for rebuilding the social contract between KMC and the people it serves, by giving both sides a shared, factual view of what is happening on the ground.

4. Platform Architecture

DevTrack is built on a modern, scalable, open-source stack. The architecture is designed to be reproducible by any municipality in Nepal — the same codebase, the same data model, the same public forum — so that DevTrack can scale from one city to all 753 local governments without rebuilding from scratch.

4.1 Technology Stack

Layer	Technology	Why This Choice
Frontend Framework	Next.js 16 (App Router, React Server Components)	Server-side rendering for fast page loads, SEO-friendly project pages, and a modern React 19 foundation.

Layer	Technology	Why This Choice
Language	TypeScript 5 (strict)	Type safety catches bugs at compile time, makes the codebase maintainable for future contributors.
Styling	Tailwind CSS 4 + shadcn/ui (New York style)	Utility-first CSS for rapid iteration, accessible component library for consistent UX.
Database	Prisma ORM + SQLite (dev) / PostgreSQL (prod)	Type-safe database access, schema-first design, easy migration from SQLite to Postgres for production.
Authentication	NextAuth.js (credentials provider in beta, Nagarik App in production)	Flexible auth that supports Nepal's national digital ID system when integrated.
State Management	Zustand (client) + TanStack Query (server)	Lightweight, predictable state without boilerplate.
Visualization	Recharts + custom SVG components	Interactive budget breakdowns, progress bars, milestone timelines.
Typography	Space Grotesk (display) + Inter (body) + JetBrains Mono (numbers)	Modern, technical aesthetic that signals "serious civic tech" rather than "government website."
AI Integration	z-ai-web-dev-sdk (planned for v1.1)	Auto-summarize 800-word project descriptions into 2-sentence plain-language overviews for non-technical citizens.

4.2 Data Model Overview

DevTrack's Prisma schema defines 14 interconnected models that capture the full lifecycle of a development project — from planning through completion, with full budget tracking, milestone management, official accountability, citizen discussion, and feedback collection.

Model	Purpose	Key Fields
User	Every registered citizen, expert, official, or admin	email, name, role (CITIZEN/EXPERT/OFFICIAL/ADMIN), expertise, wardId, avatarColor
Ward	Kathmandu's 32 administrative wards with geo-coordinates	number (1-32), name, district, population, area, latitude, longitude
Category	The 8 development sectors	slug, name, description, icon, color
Project	Every KMC development project	title, slug, summary, description, categoryId, wardId, status, priority, budgetAllocated, budgetSpent, progress, dates, contractor, consultant, fundingSource, lat/lng, featured
Milestone	Project milestones with due dates and completion tracking	projectId, title, description, dueDate, completedAt, status (PENDING/DONE/OVERDUE)
BudgetLine	Line-item budget breakdown per project	projectId, label, amount, spent
ProjectUpdate	Official announcements from KMC project teams	projectId, authorId, title, body, type (PROGRESS/BUDGET/DELAY/COMPLETION/ISSUE)
Official	Elected and appointed officials with public contact info	name, title, portfolio, wardId, phone, email, officeAddress, surgeryHours, party, termStart
ForumThread	Public discussion threads (project-specific or city-wide)	projectId (nullable), title, body, authorId, pinned
Comment	Threaded replies in forum threads	threadId, parentId (for nesting), authorId, body, isOfficial
Vote	Upvote/downvote on comments	commentId, userId, value (+1 or -1)
TrackedProject	User's tracked projects (for notifications)	userId, projectId

Model	Purpose	Key Fields
Notification	User notifications for project updates and replies	userId, title, body, link, read
BetaFeedback	Public beta feedback wall submissions	userId, authorName, category (BUG/FEATURE/IDEA/PRAISE), body, rating (1-5)

4.3 API Surface

DevTrack exposes 16 REST API endpoints under /api, all returning JSON. Every endpoint is designed for the SPA frontend but is also usable by external apps — journalists, researchers, or future mobile clients.

Endpoint	Method	Purpose
/api/projects	GET	List all projects with filters: categoryId, wardId, status, search, featured, limit
/api/projects/[slug]	GET	Full project detail with milestones, budget lines, updates, and forum threads
/api/projects/track	GET, POST	List user's tracked projects; toggle tracking on a project
/api/categories	GET	List all 8 development sectors
/api/wards	GET	List all 32 KMC wards with population and geo-data
/api/officials	GET	List all officials with contact info and ward assignments
/api/forum/threads	GET, POST	List forum threads (filterable by projectId); create a new thread
/api/forum/comments	POST	Post a reply in a forum thread (auto-tags as Official if author is OFFICIAL role)
/api/comments/vote	POST	Upvote or downvote a comment (toggle behavior)
/api/beta	GET, POST	List beta feedback wall; submit new feedback
/api/auth/login	POST	Sign in with email (and optional name for new users)

Endpoint	Method	Purpose
/api/auth/me	GET	Get current user session
/api/auth/logout	POST	Sign out
/api/stats	GET	Aggregate platform statistics (project counts, budget totals, status breakdown, category breakdown)
/api/notifications	GET	List current user's notifications
/api/route	GET	Health check endpoint

5. User Roles and Permissions

DevTrack supports four user roles, each with different capabilities. The role system is designed to be lightweight in the beta (anyone can sign up as a Citizen) and tighten in production (Citizen verification via Nagrik App, Expert verification via credentials review, Official verification via KMC email domain).

Role	How You Get It	What You Can Do
CITIZEN	Self-signup with any email	View all projects and officials, post on the public forum, reply to threads, upvote/downvote comments, track projects, submit beta feedback, receive notifications
EXPERT	Apply with credentials → Admin approval	Everything a Citizen can do, plus: green Expert badge on all comments and threads, expertise field displayed publicly, comments prioritized in KMC review queue

Role	How You Get It	What You Can Do
OFFICIAL	KMC email domain verification	Everything a Citizen can do, plus: blue Official Response badge on all comments and threads, official replies are highlighted with emerald glow on the forum, can post Project Updates on assigned projects
ADMIN	Granted by existing Admin	Everything an Official can do, plus: approve/reject Expert applications, moderate forum content, post Project Updates on any project, manage officials directory, view all user accounts

5.1 Demo Accounts (Beta)

For the public beta, DevTrack ships with 11 pre-seeded user accounts that let anyone explore the platform without signing up. The AuthModal's "Quick Login" section lists all demo accounts with one-click access.

Email	Name	Role	Ward
admin@devtrack.gov.np	DevTrack Admin	ADMIN	—
mayor@kmc.gov.np	Bidya Sundar Shakya (Demo)	OFFICIAL	—
sushil@devtrack.np	Sushil Bhattarai	EXPERT	Ward 5
kristina@devtrack.np	Kristina Khanal	EXPERT	Ward 12
anil@expert.np	Er. Anil K.C.	EXPERT	Ward 1
rajesh@citizen.np	Rajesh Maharjan	CITIZEN	Ward 22
sita@citizen.np	Sita Tamang	CITIZEN	Ward 7
sugam@devtrack.np	Sugam Parajuli	CITIZEN	Ward 15
preeti@devtrack.np	Preeti Pantha	CITIZEN	Ward 3
tushar@devtrack.np	Tushar Shah	CITIZEN	Ward 10
sampada@devtrack.np	Sampada Koirala	CITIZEN	Ward 20

6. Home Page — The Mission Front Door

The DevTrack home page is designed to make the platform's mission instantly clear to any visitor — citizen, journalist, official, or curious observer. The page uses a bento-grid layout with a dark "command center" aesthetic: an animated mesh gradient background, a 48px technical grid overlay, glassmorphic cards with backdrop blur, and a marquee ticker of all 15 project titles scrolling across the hero.

6.1 Hero Section

The hero opens with a pulse-dot "LIVE · KMC" indicator and a "For the people of Nepal" badge, immediately signaling that this is a live, civic platform. The main headline reads "Every rupee tracked. Every promise accountable. Every voice heard." — each line styled with a different accent (gradient emerald for "tracked," italic mint for "accountable," lime for "heard") to emphasize the three core promises.

Below the headline, three primary call-to-action buttons guide the visitor: "Explore Projects" (primary, emerald glow), "Open Public Forum" (glass outline), and "Find Officials" (ghost). The hero closes with a 4-column bento stats grid showing the platform's live KPIs: total budget tracked (formatted in NPR with Arba/Crore/Lakh suffixes), number of active projects across 32 wards, citizens on the platform, and forum thread count.

6.2 Mission Pillars Section

Four bento cards explain the four pillars of good governance — Transparency, Accountability, Participation, and Trust — each with an icon, a numbered index (01–04), and a one-paragraph explanation. This section makes the mission tangible and aligns DevTrack with the Open Government Partnership framework.

6.3 Featured Projects

Four flagship KMC projects are showcased as larger bento cards. Each card shows the category icon and color, ward number, project number, status badge (with glow), title, summary, progress bar (gradient emerald to lime), budget vs. spent split, and the planned end date. Clicking any card opens the full project detail page.

6.4 Live Feed — Latest Project Updates

A two-column layout shows the 5 most recent project updates on the left (with category icons, update type tags like PROGRESS/DELAY/COMPLETION, timestamps in "time ago"

format, and clickable cards that jump to the source project), and the 3 most recent forum threads on the right (with author avatars, reply counts, and a "Join the conversation" button).

6.5 Eight-Sector Grid

A 4-column grid of all 8 development sectors, each card showing the sector icon, name, description, project count, and total budget. Clicking any sector filters the Projects Explorer to that category. The cards have corner-accent brackets (L-shaped emerald lines) that give a technical, blueprint feel.

6.6 Beta Call-to-Action

The home page closes with a large glowing CTA card on a mesh gradient background: "This is the beta. Your voice shapes what DevTrack becomes." Two buttons invite the visitor to share beta feedback or read the full vision on the About page.

7. Projects Explorer

The Projects Explorer is the primary discovery surface for KMC development projects. It presents all 15 seeded projects (and any future projects added by KMC) in a responsive 3-column bento grid, with powerful filtering and full-text search.

7.1 Filtering and Search

A glassmorphic filter bar at the top of the page provides four controls: a full-text search input (matches title, summary, and description), a Sector dropdown (All sectors or any of the 8 categories), a Ward dropdown (All wards or any of 32 KMC wards), and a Status dropdown (All, Planned, Ongoing, Delayed, Completed, Stalled). When any filter is active, a results counter and a "Clear" button appear below the filter bar.

7.2 Project Card

Each project card displays: category icon and color, category name, project number (e.g. #001, #002), status badge (color-coded with badge-glow shadow), title, 2-line summary, ward and priority, progress bar (gradient emerald to lime), and a footer with formatted budget and percentage spent. Clicking any card opens the Project Detail page.

7.3 Status Categories

Status	Color	Meaning
Planned	Gray	Approved and funded but not yet started; milestone due dates are in the future
Ongoing	Blue	Currently under construction or active implementation
Delayed	Red	Behind schedule; one or more milestones are overdue
Completed	Green	Finished, handed over, and (where applicable) audited
Stalled	Amber	Paused due to legal, financial, or political issues; no current activity

8. Project Detail Page

The Project Detail page is the heart of DevTrack's transparency mission. Every KMC project gets a full page with budget breakdowns, milestone timelines, official updates, accountable officials, location data, and a direct link to the public discussion. The page is designed to give a citizen everything they need to understand a project in one scroll.

8.1 Project Header

The header shows the category icon and color, category name, ward number, the project title (large display font), the summary (larger body text), and a row of badges: status (with glow), priority (Low/Medium/High/Critical), funding source (KMC, Federal, Donor, PPP, or combinations), and contractor name. Two action buttons appear on the right: "Track this project" (toggle — when tracked, becomes "Tracking — Stop notifications") and "Open public discussion" (jumps to the project's forum thread).

8.2 Stats Row

Four stat boxes summarize the project at a glance: Budget Allocated (formatted NPR with full amount in sub-text), Budget Spent (with percentage of allocated), Milestones (completed/total, with overdue count), and Planned End Date (with start date). Each box has a category-colored icon, mono-font label, and large ticker-style number.

8.3 About This Project

The full project description, written in plain language (targeted at a non-technical citizen audience). Descriptions explain what the project is, why it matters, who benefits, what concerns have been raised, and what the expected outcomes are. For example, the Ring Road Widening project describes the 6.2 km stretch, the 8-lane design with cycle lane, the 4 skywalks, the joint funding with China Railway, and the concern about 87 trees being felled.

8.4 Budget Breakdown

The budget section starts with an overall utilization bar (gradient emerald → bright → lime) showing the total spent vs. allocated. Below that, each budget line item gets its own row: label (Materials, Labor, Equipment, Land Compensation, Overhead), spent amount, total amount, percentage, and a progress bar. This lets citizens see exactly where the money is going — not just the total.

8.5 Timeline and Milestones

A vertical timeline shows every milestone with an icon (green check for DONE, red alert for OVERDUE, gray circle for PENDING), the milestone title, due date, and completion date if applicable. Overdue milestones get a red OVERDUE badge. The timeline is the single most powerful accountability tool on DevTrack — when a milestone is overdue, every citizen can see it, and every official knows they can see it.

8.6 Project Updates

Official announcements from KMC project teams are listed in reverse chronological order. Each update has a type tag (PROGRESS, BUDGET, DELAY, COMPLETION, ISSUE — each color-coded), timestamp, author name, title, and body. Updates can be posted by OFFICIAL and ADMIN role users. Examples in the beta include "North carriageway opened to traffic," "MEP delay — steel shortage," "NICU incubators tender awarded," and "Bootcamp completed — 73% placement rate."

8.7 Project Facts Sidebar

A sidebar card lists the key facts: Contractor, Consultant, Funding source, Start date, Planned end, Actual end, and Ward. This is the quick-reference panel for journalists and researchers who need the basic facts without reading the full description.

8.8 Location Preview

For projects with geo-coordinates, a location card shows a stylized map preview with the project's latitude and longitude. The beta uses a static grid-overlay representation; v2.0 will

integrate Leaflet with Kathmandu ward GeoJSON boundaries for a true interactive map with project pins.

8.9 Public Discussion Link

If the project has an associated forum thread, a card shows the comment count and a button to open the discussion. This connects the factual project data with the citizen conversation about it.

9. Public Forum

The Public Forum is DevTrack's participation engine. Every project can have a dedicated discussion thread, and there is also a city-wide forum for general KMC topics. The forum supports threaded replies, upvote/downvote, expert and official badges, and pinned threads.

9.1 Thread Structure

A forum thread consists of: author avatar (with role-colored border), author name, timestamp, role badges (Expert, Official, Admin), expertise field (if Expert), thread title, thread body, and a comments section. Pinned threads (set by Admin) get a red PINNED badge and appear first. Project-tagged threads show the project title as a colored badge linking back to the project.

9.2 Comments and Replies

Each thread can have unlimited comments. Comments display author avatar, name, timestamp, role badges, and the comment body. Official responses (from OFFICIAL or ADMIN role users) get a special emerald-glow background and a blue "Official Response" badge — making it easy for citizens to scan a thread and see what their government has actually said.

9.3 Voting

Every comment has an upvote and downvote button. Clicking upvote adds +1 to the comment's score and highlights the thumbs-up icon in emerald. Clicking downvote adds -1 and highlights the thumbs-down in red. Clicking the same vote again removes it (toggle behavior). Voting helps surface the most useful comments — citizens can sort by score to find the highest-quality contributions, including expert analysis and official responses.

9.4 Creating a New Thread

Signed-in users can click "New thread" to open a form with title and body fields. The thread is immediately published to the forum. In the beta, there is no moderation queue — threads are live instantly. In production, an optional moderation layer will be available for ADMINS to flag or remove inappropriate content.

9.5 Forum Examples in the Beta

The beta ships with 5 seeded forum threads: 4 project-specific threads (Ring Road Widening, Bagmati Cleanup, TUTH Pediatric Wing, Indra Chowk Pedestrian Zone) and 1 city-wide thread ("How should KMC prioritize projects for the FY 2025-26 budget?"). Each project thread includes sample citizen concerns, expert analysis, and official responses — demonstrating the full conversation flow.

Sample Conversation

Sita Tamang (Citizen, Ward 7): "The night-time construction noise after 10 PM is affecting my children's sleep." Er. Anil K.C. (Expert): "From a structural engineering perspective, the tree felling could increase urban heat island effect by 1.5–2°C. Strongly recommend the 870 compensatory saplings be planted within 6 months, not the 2-year window." Mayor Bidya Sundar Shakya (Official Response): "Thank you Sita-ji and Er. Anil. I have instructed the Infrastructure Division to (1) enforce 9 PM stop-work on weekdays, (2) advance the plantation timeline to within 12 months, and (3) prioritise native shade species."

10. Officials Directory — Direct Contact

The Officials Directory is DevTrack's answer to the question every citizen has asked at least once: "Who do I even call about this?" Every official listed has a public phone number, email, office address, surgery hours (public meeting times), party affiliation, and ward assignment. Mayor and Deputy Mayor cards get an emerald-border highlight.

10.1 Search and Filter

A search input matches name, title, or portfolio. Three filter buttons let visitors narrow by role: "Mayor & Deputy" (executive), "Ward Chairs" (local), or "Dept. Chiefs" (appointed department heads). The default view shows all officials.

10.2 Official Card

Each card shows: avatar (initials in colored circle, with emerald border for Mayor/Deputy), name, title badge (emerald-filled for executive, emerald-outline for others), party badge (if any), portfolio description, and a contact list: phone (clickable tel: link), email (clickable mailto: link), office address, surgery hours, represented ward, and term start date.

10.3 Officials in the Beta

Name	Title	Portfolio	Surgery Hours
Bidya Sundar Shakya	Mayor	Chief Executive of KMC, all wards	Every Sunday 10:00–12:00
Hari Prabha Khadgi	Deputy Mayor	Social development, women & children	Every Monday 11:00–13:00
Ishwor Man Dangol	Ward Chair	Ward 1 — Administration & local disputes	Sun–Fri 10:00–12:00
Suman Maharjan	Ward Chair	Ward 5 — Heritage & tourism	Sun–Fri 10:00–12:00
Bhakta Ratna Tuladhar	Ward Chair	Ward 20 — Urban planning	Sun–Fri 11:00–13:00
Gyan Maya Lama	Ward Chair	Ward 22 — Water & sanitation	Sun–Fri 10:00–12:00
Rabin Man Shrestha	Chief, Environment Dept.	Air quality, Bagmati cleanup, solid waste	Sunday 13:00–15:00
Sarbendra Khanal	Chief, Infrastructure Dept.	Roads, drainage, street lighting	Tuesday 11:00–13:00

11. Citizen Dashboard

The Citizen Dashboard is the signed-in user's personal command center. It shows their tracked projects, their ward's projects, notifications, and quick-action links. The dashboard is role-aware — a Citizen sees their tracked projects, an Expert sees the same plus their expert badge, and an Official sees the same plus the ability to post project updates.

11.1 Profile Header

A glassmorphic card with mesh-gradient background shows the user's avatar (with emerald border), name, role badge (Expert, Official, or Admin), email, expertise (if Expert), and ward assignment. A "Sign out" button appears in the top-right.

11.2 Tracked Projects

Lists every project the user is tracking, with a status dot (color-coded by project status), title, category, budget, progress percentage, and a mini progress bar. Clicking any tracked project opens its detail page. If the user is not tracking any projects, an empty state with an "Explore projects" button appears.

11.3 My Ward

If the user has a ward assignment, this card shows up to 4 projects in their ward, with title, status, and budget. Clicking any opens the project detail. This feature makes DevTrack personally relevant — citizens see what is happening in their immediate neighborhood.

11.4 Notifications

A sidebar card lists the user's notifications — project status changes, official replies to their comments, milestone completions, budget overruns. Each notification has a title, body, and timestamp. In the beta, notifications are seeded for the Admin account; in production, notifications are auto-generated by the system when tracked projects update.

11.5 Quick Actions

Four ghost buttons link to the most common destinations: Open forum, Browse projects, Find officials, Give beta feedback.

12. Beta Feedback Wall

The Beta Feedback Wall is DevTrack's mechanism for letting citizens shape the platform itself. Every comment, bug report, feature request, and idea submitted here goes directly to the DevTrack team and to KMC leadership. The wall is public — anyone can see what others have said, with star ratings and category tags.

12.1 Submit Form

A sticky form on the left side of the page collects: your name (defaulting to the signed-in user's name), category (Feature request, Bug report, Idea, or Praise), feedback body, and a 1–5 star rating. The submit button posts the feedback to the wall instantly.

12.2 Feedback Categories

Category	Color	Use Case
Feature request	Emerald	You want DevTrack to do something it doesn't do yet (e.g., "Add a Nepali language toggle")
Bug report	Red	Something is broken (e.g., "Filter by ward shows wrong projects when category is also selected")
Idea	Lime	A suggestion that's not a specific feature (e.g., "Add an Expert Verified badge on comments approved by 3+ domain experts")
Praise	Mint	You love something (e.g., "Finally a place where my voice reaches the Mayor!")

12.3 Feedback Wall

The right side of the page shows all submitted feedback in reverse chronological order. Each item shows the author's avatar, name, category badge (color-coded), timestamp, star rating (5 stars, filled for the rating), and the feedback body. The wall is read-only — no replies or voting in the beta, to keep the focus on collecting raw input.

12.4 Average Rating

The page header shows the average star rating across all feedback, giving the team a quick pulse on overall satisfaction. In the beta, the average is 4.5/5 across 6 responses.

13. About / Vision Page

The About page tells the full DevTrack story: who built it, why, what problem it solves, how it aligns with global frameworks, and where it's going. The page is designed for stakeholders evaluating DevTrack — funders, government partners, OGP reviewers, journalists writing about civic tech in Nepal.

13.1 Mission Statement

A glowing emerald card with mesh-gradient background contains the DevTrack mission quote: "DevTrack is for the people. It exists so that every Nepali — in Kathmandu today, in all of Nepal tomorrow — can see what their government is doing, question it, and shape it. We build it in the open, with citizens, for citizens."

13.2 Problem, Solution, Frameworks, Beyond Kathmandu

Four sections explain: the problem DevTrack solves (transparency gap in KMC projects), the solution it provides (centralized platform with 8 sectors, full budget breakdowns, named officials, public forum), the global frameworks it aligns with (OGP, NeGIF, World Bank Digital Dividends 2016), and the vision for scaling beyond Kathmandu to all 753 local governments of Nepal.

13.3 Six Goals

A numbered list of 6 goals: (1) Enhance community engagement, (2) Improve financial accountability, (3) Ensure accessible information, (4) Create a public forum for citizen voices, (5) Prioritise expert opinions, (6) Position KMC for OGP membership.

13.4 The Team

Six student researchers from across Nepal, guided by one mentor (Shubham Jha, Franklin & Marshall College) and one peer mentor (Prashim Timsina, Trinity Int'l College). Each team member has a card with avatar, name, role badge, and bio. The page closes with a Sustainability section explaining how DevTrack will become self-sustaining as KMC adopts it as the primary project information platform.

14. Design System

DevTrack's visual identity is built on a dark "command center" aesthetic with emerald green as the primary accent. The design signals "serious civic tech" rather than "government website" — modern, technical, trustworthy.

14.1 Color Palette

Token	Hex	Usage
Background	#0C1F1A (oklch 0.10 0.012 165)	Near-black with subtle green tint — the base canvas
Card	#13201C	Slightly lighter dark for glassmorphic cards
Emerald Glow	#10B981	Primary accent — buttons, active states, key data
Emerald Bright	#34D399	Secondary accent — labels, highlights
Lime Accent	#A3E635	Tertiary accent — ratings, success states, CTAs
Mint Mist	#6EE7B7	Quaternary accent — italic text, soft highlights
Foreground	#F7FBF8	Near-white body text
Muted Foreground	#A0B5AC	Captions, metadata, mono labels

14.2 Typography

Three font families work together: Space Grotesk for display headlines (geometric, modern, tech-feel), Inter for body text (highly readable at small sizes), and JetBrains Mono for numbers, labels, and code-style annotations (e.g., "// LIVE FEED", "// 8 SECTORS").

14.3 Visual Effects

- Glassmorphism — backdrop-blur(16px) + saturate(180%) on all cards, dropdowns, and modals.
- Animated mesh gradient — radial gradients at 4 positions, shifting over 14s, creating a living background.
- Grid overlay — 48px grid lines at 6% opacity, giving a technical blueprint feel.

- Glow effects — emerald box-shadows on buttons and CTAs (glow-emerald, glow-emerald-strong).
- Gradient text — emerald → mint → lime gradient on key headlines (text-gradient-emerald).
- Bento cards — lift-on-hover with mouse-tracking radial gradient, corner-accent brackets.
- Marquee — 40s infinite scroll of all 15 project titles across the hero.
- Pulse dot — animated ping ring on the "LIVE · KMC" indicator.

14.4 Logo

Custom SVG logo: a hexagonal outer ring (representing structure and governance), an inner D-mark (for DevTrack), and a pulse line (representing the live, beating heart of civic engagement). The gradient runs from mint (#6EE7B7) through emerald (#10B981) to lime (#A3E635), matching the platform's accent palette. The logo scales cleanly from 28px (header) to 48px (large).

15. Seeded Project Portfolio

The DevTrack beta ships with 15 flagship KMC development projects across all 8 categories. These projects are realistic, mission-driven examples that demonstrate every feature of the platform — budget breakdowns, milestone timelines, official updates, forum discussions, and accountable officials.

Project	Category	Ward	Status	Budget (NPR)
Ring Road Widening — Kalankati to Maharajgunj	Infrastructure	5	Ongoing	1.85 Arba
Bagmati River Cleanup — Phase 4	Environment	11	Ongoing	42 Crore
Tribhuvan University Teaching Hospital — Pediatric Wing	Healthcare	4	Delayed	98 Crore

Project	Category	Ward	Status	Budget (NPR)
Shree Janasewa Secondary School — Earthquake Retrofit	Education	7	Ongoing	14.5 Crore
Asan Bazaar Heritage Market Modernization	Urban Development	5	Planned	22 Crore
Senior Citizen Allowance Digital Disbursement 2024	Citizen Benefits	1	Completed	1.82 Arba
Kathmandu Air Quality Monitoring Network — 12 Stations	Environment	15	Ongoing	9.5 Crore
Indra Chowk Pedestrian-Only Zone Pilot	Policy	5	Planned	2.8 Crore
Balaju Ward 22 Stormwater Drainage System	Infrastructure	22	Ongoing	34 Crore
Kathmandu Youth Tech Bootcamp 2024 — 1,000 Trainees	Economy	20	Completed	8.5 Crore
Solar Rooftop at 4 KMC Health Posts	Environment	5	Ongoing	4.8 Crore
New Road Smart Parking — 380 bays	Urban Development	1	Ongoing	6.5 Crore
Ward 20 — 6 Public Toilets with Disabled Access	Citizen Benefits	20	Ongoing	1.8 Crore

Project	Category	Ward	Status	Budget (NPR)
KMC e-Sewa — Unified Citizen Services Portal	Policy	1	Ongoing	22 Crore
KMC Disaster Early Warning & Response System	Policy	1	Planned	17.5 Crore

15.1 Total Platform Metrics

Metric	Value
Total projects	15
Total budget tracked	NPR 6.51 Arba (≈ \$48.9 million USD)
Total budget spent	NPR 4.44 Arba (68% deployment rate)
Total officials	8 (Mayor, Deputy Mayor, 4 Ward Chairs, 2 Dept. Chiefs)
Total wards	32 (full KMC coverage)
Total registered citizens	6 seeded + 4,287 simulated = 4,293
Total forum threads	5 (4 project-specific + 1 city-wide)
Total beta feedback	6 items, average rating 4.5/5
Total demo user accounts	11 (1 Admin, 1 Official, 3 Experts, 6 Citizens)

16. Roadmap — What's Next for DevTrack

DevTrack launches tonight as a public beta for Kathmandu Metropolitan City. The roadmap below outlines the features planned for v1.1 through v2.0, based on the team's vision and the beta feedback received so far.

16.1 v1.1 — Immediate Priorities (Next 30 Days)

- Nepali language toggle — full localization of the UI into Nepali (Devanagari script), since most KMC citizens prefer Nepali over English.

- AI project summaries — auto-generate 2-sentence plain-language overviews of 800-word project descriptions using z-ai-web-dev-sdk, for non-technical citizens.
- Expert Verified badge — comments approved by 3+ domain experts get a special badge, increasing their visibility in KMC's review queue.
- Search across all content — unified search bar that queries projects, forum threads, officials, and beta feedback simultaneously.

16.2 v1.2 — Data and Verification (60–90 Days)

- Real KMC data scraping pipeline — automated scripts that pull project data from official KMC sources (PPMO, KMC website, Nagarik App) on a weekly schedule.
- Nagarik App SSO — replace the beta's simplified auth with Nepal's national digital ID system, including biometric verification for Official accounts.
- Email and SMS notifications — when a tracked project updates, citizens receive an email or SMS (via SparrowSMS Nepal) in their preferred language.
- Photo evidence gallery — citizens and officials can upload photos of project sites, geotagged and timestamped, creating a visual record over time.

16.3 v2.0 — Scale and Integration (6–12 Months)

- Interactive Leaflet map with Kathmandu ward GeoJSON boundaries, project pins, and ward-level clustering.
- Scale to all 753 local governments of Nepal — same data model, same public forum, same accountability framework, deployed per-municipality.
- Mobile apps (Android and iOS) — React Native apps with offline-first design for areas with poor connectivity.
- Official KMC integration — DevTrack becomes the canonical source of project information, with KMC staff posting updates directly through a dedicated official portal.
- Open Government Partnership submission — formal OGP commitment package using DevTrack as the centerpiece of KMC's transparency pledge.

17. Technical Implementation Details

17.1 File Structure

DevTrack follows the Next.js 16 App Router convention. All source code lives under `/src`. The 9 view components are in `/src/components/devtrack/`. The 16 API routes are in

/src/app/api/. The Prisma schema is in /prisma/schema.prisma. The seed script is in /scripts/seed.ts.

17.2 Database Seeding

The seed script (scripts/seed.ts) populates the database with the full mission-driven KMC dataset: 8 categories, 32 wards with real Kathmandu coordinates, 11 demo users across all 4 roles, 8 officials with contact info, 15 flagship projects with milestones/budget lines/updates/forum threads, 6 beta feedback items, and 1 admin notification. Running the seed takes ~3 seconds and produces a fully functional demo environment.

17.3 Authentication Flow

The beta uses a simplified credentials-based auth: the /api/auth/login endpoint accepts an email (and optional name for new users), looks up the user in the database, creates them if they don't exist, and sets a devtrack_uid HTTP-only cookie. The /api/auth/me endpoint reads the cookie and returns the current user. The /api/auth/logout endpoint clears the cookie. In production, this will be replaced by NextAuth.js with the Nagarik App OAuth provider.

17.4 Performance

DevTrack uses Next.js 16 with Turbopack for fast dev compilation (~600ms cold start, ~25ms warm page loads). The home page makes 7 parallel API calls on mount (projects, categories, wards, officials, forum threads, beta feedback, stats) and shows a loading spinner until all resolve. All API responses are cached in React state, so navigation between views is instant after the initial load.

17.5 Responsive Design

Every page is mobile-first responsive. The header collapses to a hamburger menu (Sheet component) on screens below md (768px). The bento grids reflow from 4 columns to 2 to 1. The filter bar on Projects Explorer stacks vertically on mobile. All touch targets are at least 44px. The platform has been tested at 375px (iPhone SE), 768px (iPad), and 1280px+ (desktop).

18. Alignment with Global Frameworks

18.1 Open Government Partnership (OGP)

DevTrack directly supports all four OGP core principles: Transparency (public project data), Civic Participation (public forum), Public Accountability (named officials, milestone tracking), and Technology & Innovation (modern web platform). KMC's adoption of DevTrack would form the centerpiece of an OGP action plan commitment, positioning Kathmandu as a leader in South Asian civic tech.

18.2 Nepal e-Government Interoperability Framework (NeGIF)

DevTrack's API design follows NeGIF's interoperability standards, enabling future integration with other Nepali government systems — the Nagarik App for identity, the PPMO portal for procurement data, and NEA/Nepal Telecom for utility service integration.

18.3 World Bank Digital Dividends (2016)

The World Bank's 2016 World Development Report on Digital Dividends identified three analog complements for digital governance to deliver value: regulations, skills, and institutions. DevTrack addresses all three: it provides a regulatory framework (public project disclosure), builds citizen skills (digital literacy through the forum), and strengthens institutions (KMC's accountability mechanisms).

19. Conclusion

DevTrack is more than a website. It is a statement that citizens of Kathmandu deserve the same transparency, accountability, and voice that citizens of Stockholm, Seoul, or São Paulo take for granted. It is built by Nepali students, for Nepali citizens, with Nepali government data, aligned with global open government standards.

The beta launches tonight with 15 flagship projects, 32 wards, 8 officials with direct contact info, a public forum with seeded citizen-expert-official conversations, and a beta feedback wall that lets every visitor shape what DevTrack becomes. Every feature described in this document is live, tested, and ready for citizens to use.

The mission is simple: every rupee tracked, every promise accountable, every voice heard. The platform is the tool. The people of Nepal are the purpose.

Get Involved

Visit DevTrack at the preview link. Sign in as the Mayor (one click) and post an official response. Sign in as an Expert and contribute domain analysis. Sign in as a Citizen and share your concerns. Submit beta feedback. Track a project in your ward. This is your platform. Use it. Shape it. Make it yours.